Inference as Consolidation: Continual Learning with No Offline Phase by Replay in the Silent Degrees of Freedom

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Abstract

An agent that keeps learning in deployment may not be able to afford dedicated consolidation downtime, yet replay-based continual learning almost always relies on offline rehearsal phases or on interleaving replayed samples with the incoming stream. Brains face the same constraint and solve part of it with *local sleep*: brief, use-dependent off-periods of individual circuits during wakefulness. We ask whether a network trained by local, biologically constrained rules can consolidate *during inference*, with no offline phase at all. We introduce (i) an *isolation* rule that confines replay updates to hidden synapses invisible to the current input under k-winner-take-all dynamics — exact for silent presynaptic and suppressed units, near-exact elsewhere by measurement — (ii) a *refractory rotation* rule under which units that have just fired sit out the next competition, widening the isolated channel, and (iii) internal control signals: a unit-level homeostatic pressure that triggers replay bursts and a relative-novelty gate. The construction inverts the usual direction of non-interfering continual learning: the hidden computation is held invariant — exactly on the proven channel, near-exactly end to end — while past memories are consolidated through degrees of freedom the current batch leaves unused. On class-incremental split-MNIST the system reaches $92.7 \pm 0.3\%$ with no offline phase, above the best offline-rehearsal schedule ($90.7 \pm 0.4\%$), at $2.4\times$ its replay samples; at a matched $1.2\times$ budget, eight-sample micro-batches still reach $91.8 \pm 0.3\%$, where unmasked replay and offline rehearsal degrade sharply; the ordering transfers to split CIFAR-10. A drive-referenced controller for the rule's synaptic decay replaces per-dataset tuning with one dataset-independent ratio target and, with locally learned skip synapses, lets a five-hidden-layer stack match the two-layer record where every tested fixed decay collapses.

1 Introduction

Foundation-model and embodied agents are increasingly expected to keep learning after deployment, from streams that are non-stationary and never task-labelled. The dominant tool against forgetting is replay from an episodic memory, and nearly every replay-based method consumes it in one of two ways: in a dedicated offline phase — a "night" during which the agent stops acting and rehearses — or by interleaving replayed samples with the live stream. Both cost an agent. An offline phase is downtime during which the system is unavailable or silently changing; interleaved replay perturbs the computation currently serving the input, and grows fragile as replay batches shrink toward what an online budget allows.

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Biology suggests a third option. Sleep-like off-periods occur *locally*, in individual cortical circuits of awake animals [Vyazovskiy et al., 2011], and inducing such ON/OFF alternation in one hemisphere of awake mice discharges sleep pressure and rescues memory consolidation, whereas an equal tonic reduction of firing does not [Driessen et al., 2026]. Some core functions of sleep can thus be discharged locally during wakefulness. We take this as an algorithmic motivation — not as a claim about where biological consolidation occurs — to ask whether the computational degrees of freedom a learner is not using can host consolidation while it acts, and what that costs.

We study the question in a deliberately constrained learner — a two-hidden-layer network trained end to end by local, biologically motivated rules (no weight transport, bounded burst-like error channel, Dale's law, sparse population codes, sparse distance-dependent wiring, single forward–feedback sweep) — because those constraints expose exactly which degrees of freedom the current input does and does not use. The contributions are:

- **Isolated replay during inference.** A synapse may take a replay update only if its presynaptic unit is silent for the current input or its postsynaptic unit is asleep. Under k-WTA dynamics this leaves the hidden computation exactly invariant for silent-presynaptic and suppressed units (Propositions 1–2); we measure how nearly it holds elsewhere and end to end. Isolation makes micro-batch replay affordable: accuracy is flat from 256 down to 8 replayed samples per step, where unmasked and offline schedules collapse.
- **Refractory rotation.** Units that fired on the current batch sit out the next competition. This alternation — the algorithmic counterpart of the ON/OFF requirement in Driessen et al. [2026] — widens the consolidable channel from ≈ 0.25 to 0.53 of the replay-active units and is worth $+3.9$ sequential points; a tonic reduction of the same units' gain, like the biological control, is not.
- **Internal triggers.** Replay is triggered by a unit-level homeostatic pressure; the tested surprise trigger fails during wake, whereas offline rehearsal is best timed by novelty in our sweeps. A relative-novelty gate on rotation is scale-free where absolute thresholds fail.
- **One ratio target instead of per-dataset decay tuning, and depth.** A drive-referenced controller for the rule's synaptic decay removes its last per-dataset constant and, with locally learned skip synapses, lets a five-hidden-layer stack match the two-layer record on both axes where every tested fixed decay collapses to chance.

Everything is evaluated on two axes at once — the class-incremental stream and the identical learner on the i.i.d. shuffle of the same data — because consolidation during inference must not damage ordinary learning. The scale is small (MLPs on MNIST/CIFAR variants); the object of study is the mechanism, whose assumptions (sparse codes, a local error signal) are the ones a larger system would have to satisfy to inherit the guarantee.

2 Related work

Sleep-inspired consolidation. The Bazhenov line implements sleep as a global offline phase driven by noise under local plasticity [Tadros et al., 2022, Bazhenov et al., 2026, Golden et al., 2022, Kubo et al., 2025]; wake–sleep frameworks with NREM/REM stages [Sorrenti et al., 2025, Robinson et al., 2022] and systems such as SIESTA [Harun et al., 2023] likewise schedule consolidation offline. In all of these sleep is global, offline and clocked; none consolidates in currently unused units during inference.

Gating and history-dependent competition. Context-dependent gating activates a subnetwork per task from task identity [Masse et al., 2018]; learned gates [Tilley et al., 2023], dendritic context [Iyer et al., 2022], task-free sparsification [Lässig et al., 2023] and recovered functional masks [McKee et al., 2026] derive the gate from the input. Refractory competition has classical precedents [Kaski and Kohonen, 1994, Maeda and Miyajima, 1999], and spiking continual learners raise firing thresholds with use [Shen et al., 2024]. Our rotation differs in purpose: recently active units are benched for one competition to enlarge a *replay channel*, not to allocate resources across tasks.

Null-space and parameter isolation. A mature line protects *previous* tasks: gradient or activation-subspace projections [Farajtabar et al., 2020, Saha et al., 2021], feature-covariance null spaces [Wang et al., 2021], k-winner sparsity with such projections [Abbasi et al., 2022], frozen subnetworks [Golkar et al., 2019, Mallya and Lazebnik, 2018]. We invert the protected direction: the ongoing computation is held invariant while old-memory replay is written into degrees of freedom the current sparse support exposes as invisible — no stored bases or task masks, and the "null space" changes every batch for free. McKee et al. [2026] and Bricken et al. [2023] note that optimiser state can break such guarantees; we adopt mask-confined optimiser state rather than claiming it. Complementary-learning-system methods [Arani et al., 2022, Pham et al., 2021] raise the value of each replayed sample but still replay at every step; learned replay scheduling [Klasson et al., 2023] is related to our trigger study.

3 Method

Setting. A stream of labelled examples is presented in T contiguous tasks with disjoint class sets; no task identity is available at test time; a single shared C-way readout is evaluated on all classes after the last task (class-incremental learning). We report final accuracy A_{seq}, forgetting F, the accuracy A_{iid} of the identical learner and budget on the i.i.d. shuffle (the *static* axis), the replay cost R in samples, and an arithmetic-energy proxy E; the learner is confined to the cortical constraint set $\mathcal{B}$ above and to an episodic memory of K raw examples with reservoir writing [Vitter, 1985]. The question is whether the offline term of consolidation can be removed entirely without losing A_{seq}, and at what price in A_{iid} and R.

Base learner. Layer activities are $a^0 = x$ and, for $\ell = 1, \ldots, L$,

$$z^\ell = W^\ell a^{\ell-1} + b^\ell, \qquad a^\ell = \Phi_k\big(\sigma(z^\ell) \odot (1 - s^\ell)\big), \tag{1}$$

where σ is a rectifier, $s^\ell \in \{0,1\}^{n_\ell}$ the suppression mask of the rotation rule below, and Φ_k keeps the k largest entries of each row (k-WTA); a per-unit gain exists in the implementation for homeostatic scaling but is fixed at 1 throughout. Errors come from one top-down sweep through learned feedback weights $B^\ell \in \mathbb{R}^{n_{\ell+1} \times n_\ell}$ (no weight transport),

$$\varepsilon^L = y - z^L, \qquad \varepsilon^\ell = \sigma'(z^\ell) \odot \Big((B^\ell)^\top \big[\kappa \tanh(\varepsilon^{\ell+1}/\kappa) \odot \mathbf{1}(a^{\ell+1} > 0) \big] \Big), \tag{2}$$

a bounded, event-gated burst signal in the spirit of predictive-coding and burst-dependent learners [Whittington and Bogacz, 2017, Payeur et al., 2021]. Forward and feedback weights follow the same local product with a shared decay λ (Kolen–Pollack alignment [Kolen and Pollack, 1994, Akrout et al., 2019]), $\Delta W^\ell \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda W^\ell$, $\Delta B^{\ell-1} \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda B^{\ell-1}$, under Dale's law with sign-concordant feedback. The optimiser is heavy-ball SGD — one per-synapse velocity, no second-moment state; Section 4 shows why neither Adam nor momentum-free SGD is the right choice inside the consolidation loop. The base learner costs 1.2 points against a dense backpropagation MLP on i.i.d. MNIST at $4\times$ fewer synaptic operations (Appendix B).

Isolated replay. Let $X = \{x_b\}_{b=1}^m$ be the current waking batch, $a^\ell_{i,b}$ the activity of unit i on sample b, and $\mathcal{A}^\ell(X) = \{i : \exists b,\ a^\ell_{i,b} \neq 0\}$ the awake set. During the same forward step, a replay micro-batch drawn from the buffer is applied through the synapse mask

$$M^\ell_{ij} = \mathbf{1}\big[\, j \notin \mathcal{A}^{\ell-1}(X) \ \vee \ i \notin \mathcal{A}^\ell(X) \,\big], \tag{3}$$

i.e. a synapse takes the replay update iff its presynaptic unit is silent for the current input or its postsynaptic unit is asleep. The mask applies to the *entire* replay-induced parameter change, not only to the data gradient: with heavy-ball velocity v ($\mu = 0.9$), data gradient G and decay λ, a replay step is

$$v \leftarrow M \odot (\mu v + G) + (1 - M) \odot v, \qquad W \leftarrow W + \eta\, M \odot v - \lambda\, M \odot W, \tag{4}$$

so the velocity is frozen (not zeroed) outside the mask, the decay is confined to it, and biases follow the unit mask $\mathbf{1}[i \notin \mathcal{A}^\ell(X)]$; unmasked momentum would re-inject the update into awake synapses [McKee et al., 2026, Bricken et al., 2023]. The readout row and its bias stay plastic, so old classes remain calibrated; the guarantees below concern the hidden computation.

Proposition 1 (Pre-silent updates are exactly invisible)**.** *Fix X and let $\widehat{W}^\ell = W^\ell + \Delta^\ell$ with $\Delta^\ell_{ij} = 0$ whenever $j \in \mathcal{A}^{\ell-1}(X)$. Then every hidden activity on every sample of X is unchanged for any magnitude of Δ; if the readout row satisfies the same condition, or is held fixed, the network output is unchanged as well.*

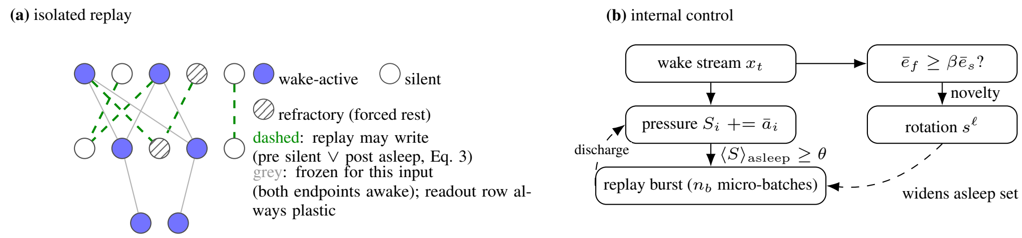

Figure 1: **Mechanism.** (a) During each waking batch, replay from the episodic buffer is written only into hidden synapses invisible to the current input — exactly so for silent presynaptic and suppressed units; refractory rotation forces recently used units into the consolidable set. (b) Unit-level homeostatic pressure triggers replay bursts; a relative-novelty gate decides when rotation runs.

Proposition 2 (Post-asleep updates are invisible under a margin)**.** *Let Δ^ℓ and Δb^ℓ additionally move synapses and biases of units $i \notin \mathcal{A}^\ell(X)$ with active presynaptic partners, and write $\delta_{i,b} = \sum_j \Delta^\ell_{ij} a^{\ell-1}_{j,b} + \Delta b^\ell_i$. If for every such unit and every sample $\sigma(z^\ell_{i,b} + \delta_{i,b}) < \tau^\ell_{k,b}$, where $\tau^\ell_{k,b}$ is the k-th largest value entering Φ_k on sample b (strict, so a tie cannot promote the unit), the hidden activities on X are unchanged.*

Corollary 1 (Suppressed units are exact for any magnitude)**.** *If $s^\ell_i = 1$ in (1), $a^\ell_{i,b} = 0$ for every sample and Proposition 2 holds with no margin condition.*

Proofs are in Appendix A. The default system does not enforce the margin, so we measured it: re-inferring every waking batch after each of its 18,760 isolated replay updates on the record configuration, the update altered the top hidden code of 0.36% of waking samples per step and changed a waking prediction for 0.27% (almost all through the plastic readout: 0.001% with the readout isolated), violating the margin for $\sim 10^{-4}$ of asleep units per step. Isolation is exact for silent-presynaptic and suppressed units and near-exact elsewhere; Corollary 1 is why rotation makes it robust.

Refractory rotation. After batch X_t, set $s^\ell_i(t{+}1) = \mathbf{1}[i \in \mathcal{A}^\ell(X_t)]$: every unit that fired sits out the next competition and is therefore asleep — consolidable, and exactly so — for X_{t+1}. Rotation widens the isolated channel ρ_ℓ (the fraction of replay-activated units that are asleep for the current input) from 0.2–0.3 to ≈ 0.53, at the price of running inference on the second-best coalition.

Internal control (Figure 1b). *When to replay:* each unit integrates its own use, $S_i \leftarrow S_i + \bar{a}_i(x_t)$ — a unit-level analogue inspired by the homeostatic Process S of sleep regulation [Borbély, 1982] — and a burst of n_b replay micro-batches fires when the mean pressure of currently asleep units exceeds θ; consolidation discharges the pressure of the units that received it. *When to rotate:* with fast and slow exponential averages of the batch error, $\bar{e}_f$ (α_f=0.1) and $\bar{e}_s$ (α_s=0.002), rotation runs while $\bar{e}_f \geq \beta\, \bar{e}_s$ ($\beta \approx$ 1.1–1.5) or while $\bar{e}_f \geq \gamma e_0$ with e_0 the chance-level error of the first 20 batches ($\gamma \approx 0.5$): the stream is new relative to the learner's own recent history, or is not yet mastered. The detector is a standard drift monitor [Ross et al., 2012], loosely inspired by cholinergic novelty signalling [Hasselmo, 2006]; what is new is what it gates; absolute thresholds cannot serve, since the converged error of a ten-class i.i.d. stream exceeds any level a two-class task dips under. At the activity fraction used throughout the gate turns out to be nearly neutral, and the default system runs rotation always on (Section 4).

Protocol. Split-MNIST (five tasks of two classes, five epochs per task) and split CIFAR-10 (same protocol, 32$\times$32 luminance), each with its i.i.d. counterpart; a third regime feeds CIFAR-10 through a frozen V1-like patch front-end (1024-D, linear probe $\approx 49\%$); split CIFAR-100 appears in Section 5 for mechanism only. Architecture d–512–256–C, k-WTA 10%, 30% distance-dependent connectivity, 80% excitatory units; waking batches of 16; buffer K=1000 (K=200 in Appendix F). Baselines: BP with experience replay (BP+ER) at equal K; the offline "night" (20 replay batches of 256 at 3$\times$ learning rate after each waking epoch, the best-performing offline schedule in our sweep) on its preferred narrower core and on our substrate; interleaved ER for the local learner; no buffer. Three seeds (six on headline rows), mean $\pm$ s.d. Energy is an idealised 45-nm FP32 arithmetic-energy proxy [Horowitz, 2014] — 0.9 pJ per FP32 accumulate (one per synaptic event) against 4.6 pJ

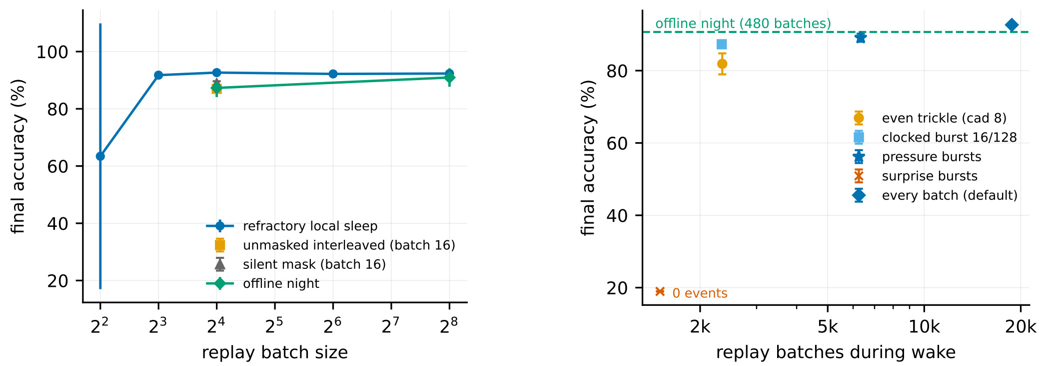

Figure 2: **Left (isolation):** split-MNIST accuracy against replay batch size: masked refractory replay is insensitive down to 8 samples; unmasked, silent-masked and offline-night replay degrade. **Right (timing):** accuracy against the number of waking replay events: trickles fail, clocked bursts recover most of the gap, pressure-triggered bursts (star) dominate at sparse budgets, surprise triggering never fires.

per multiply–accumulate — after rate-to-spike conversion with thresholds calibrated on training images [Rueckauer et al., 2017]; it excludes data movement and membrane updates. Every choice made during development used the official test sets; Appendix J replicates the headline orderings on a held-out tenth of the training data ($\rho = 0.97$, mean gap 0.9). Further protocol details are in Appendix C.

4 Results

Local sleep replaces the night. On split-MNIST at K=1000, refractory local sleep with one isolated replay micro-batch beside every waking batch reaches $\mathbf{92.7 \pm 0.3}\%$ (six seeds) with no offline phase — above the best offline night ($90.7 \pm 0.4\%$ on its preferred narrow substrate; 90.9 ± 3.2, highly variable across seeds, on ours) and far above unmasked interleaved ER at the same budget (89.7 ± 1.1) and BP+ER (89.3 ± 0.8). Adding a night on top of continuous local sleep produces no detectable improvement (92.7 ± 0.4): once consolidation runs during wake, the offline phase is redundant at this buffer size. The headline configuration replays $2.4\times$ the night's samples; at a matched $1.2\times$ budget the system still reaches 91.7–91.8 (eight-sample micro-batches every batch, or 16-sample ones every second batch; Figure 2, Appendix C).

Isolation makes micro-batch replay affordable. Figure 2 (left): masked refractory replay is flat in replay batch size from 256 down to 8 ($92.3 \pm 0.1 \to 91.8 \pm 0.3$; it breaks only at 4), while at batch 16 unmasked replay sits at 87.0 ± 1.1, the rotation-free silent mask at 88.8 ± 0.8, and the offline night falls from 90.9 to 87.3 ± 3.2 when its batches shrink to 16. A replay micro-batch is a high-variance gradient estimate; applied to the full network it perturbs the coalition currently encoding the stream and, with shared optimiser state, the perturbation compounds. Inside the exact portion of the mask each update is provably hidden-state neutral, and violations in the margin-conditioned remainder are rare in measurement, so variance accumulates only where it can do no immediate harm and is averaged over thousands of events.

Component prices (Table 1). Removing rotation costs 3.9 sequential points (88.8 ± 0.8: the pre-silent channel alone), removing isolation 5.7 with a 3.7-fold larger seed deviation (87.0 ± 1.1), and removing replay everything (19.6, the forgetting floor). The static column identifies who pays: rotation is the only statically costly component, about 1.2 points at this activity fraction (94.7 against 95.9–96.1 without it). Three replacements sharpen the construction. *Mirror isolation* — the classical protected direction with our machinery: waking updates barred from synapses that served the last replay batch, replay itself unmasked — reaches only 77.9 ± 1.6: the protected direction is a fifteen-point effect, not a framing choice. *Soft rotation* (halving rather than silencing recent winners, a tonic reduction) loses on both axes ($90.2 \pm 0.4/86.4 \pm 0.3$): all-or-none OFF periods beat turned-down gain, as in the biological comparison. *Optimiser state:* masked Adam trails heavy-ball

Table 1: **Component ablation** (512–256, k-WTA 10%, K=1000, one replay micro-batch of 16 per waking batch). Sequential = split-MNIST; static = i.i.d. MNIST. Headline rows six seeds, the rest three.

Variant	Sequential	i.i.d. (static)
full system: isolated replay + always-on rotation (heavy-ball SGD)	$\mathbf{92.7 \pm 0.3}$	94.7 ± 0.3
rotation gated by relative novelty, committed to bouts	92.3 ± 0.2	94.9 ± 0.3
rotation gated per batch	89.7 ± 1.4	95.3 ± 0.2
− rotation (pre-silent isolation only)	88.8 ± 0.8	95.9 ± 0.2
− isolation (unmasked interleaved replay)	87.0 ± 1.1	$\mathbf{96.1 \pm 0.1}$
− replay (no buffer)	19.6 ± 0.0	95.8 ± 0.2
masked Adam, moments confined to the mask	91.8 ± 0.6	94.6 ± 0.2
masked Adam, moments leaking outside the mask	92.6 ± 0.3	94.3 ± 0.3
momentum 0 (pure SGD; η and replay gain re-tuned, else chance)	89.9 ± 0.3	90.1 ± 0.2
offline night instead of concurrent replay, same substrate	90.9 ± 3.2	—
the same night on its preferred narrow substrate	90.7 ± 0.4	—
mirror direction: protect the past	77.9 ± 1.6	92.3 ± 0.2
soft rotation: gain 0.5 instead of 0	90.2 ± 0.4	86.4 ± 0.3
unmasked ER at the same budget (Adam)	89.7 ± 1.1	95.4 ± 0.3
backprop (+ER sequential; plain static), dense	89.3 ± 0.8	97.5 ± 0.0

SGD ($91.8 \pm 0.6/94.6 \pm 0.2$) because the guarantee freezes its moments outside each batch's mask, where they go stale across a unit's on/off cycles [Bricken et al., 2023]; letting the moments leak recovers most of the gap but surrenders the guarantee (92.6 ± 0.3), while a single velocity keeps both. Momentum-free SGD is not the answer either: without the velocity's noise averaging, isolated micro-batches collapse the network to chance unless the replay step is re-scaled by hand, and the best pure-SGD setting still trails by $2.8/4.6$ (Appendix H). The velocity is the least state that keeps the guarantee and the accuracy at once; a two-sided learning-rate sweep (Appendix G) shows these comparisons are plateaus, not knife-edges.

When replay must be rare: homeostatic bursts, never surprise. Figure 2 (right): at 12.5% of the replay events, a clocked burst of 16 batches every 128 retains 87.3 ± 0.2 while an even trickle at the same budget collapses (81.9 ± 2.9) — masked replay reaches only part of the synapses per event and cannot dribble. Unit-level pressure triggering does better still (89.1 ± 1.1 at a third of the events), and surprise triggering fails: the converged error never crosses the threshold, so zero replay events fire (19.0). Homeostatic triggering consistently outperforms the tested surprise trigger during wake — the opposite of the offline night, whose best trigger in our sweeps is novelty-timed.

The i.i.d. price and the activity fraction. Rotation's static price is governed by the substrate's activity fraction: at 5% it cost 1.3–3.0 points on i.i.d. MNIST and novelty or bout-committed gates were needed to refund it; at 10% the price shrinks to about a point and the gates become neutral (Table 1), because the rotation-free ceiling itself rises with the fraction. On feature CIFAR-10, relaxing the fraction improves both axes monotonically ($40.5 \to 42.7$ sequential, $39.4 \to 45.6$ static from 5 to 25%), with the local learner 1.0–1.3 points ahead of a matched-sparsity BP control at every level. Increasing the activity fraction reduced the static penalty more consistently than any tested gate (Appendices D–E).

Transfer to CIFAR-10. On raw split CIFAR-10 the ordering transfers and widens: refractory local sleep $29.5 \pm 0.8 >$ night $26.2 \pm 0.6 >$ BP+ER $25.1 \pm 1.5 >$ unmasked ER $17.0 \pm 6.1 \approx$ no buffer 16.4, despite task-active code overlap of 0.9–0.98. On the feature front-end the local ordering persists (refractory $41.5 \pm 1.1 >$ night $35.0 \pm 0.5 \gg$ unmasked 20.1 ± 7.3) while BP+ER reaches 43.2 ± 1.0 at its best learning rate — a 1.7-point lead. Transplanting the local network's ingredients into the BP net one at a time (each at its better of two learning rates) is consistent with activation sparsity accounting for that mean gap: width helps BP+ER (44.5 ± 1.0), the 30% distance-dependent wiring costs 1.2 (43.3 ± 0.9), and k-WTA costs the rest (40.5 ± 0.7), leaving BP+ER at or below the local learner on the identical substrate. The rotation's static effect on raw CIFAR is inconclusive under the available seeds (27.6 ± 0.4 vs. 29.5 ± 2.8).

Table 2: **One dataset-independent ratio target replaces per-dataset decay tuning.** Sequential / static accuracy under the tuned fixed decay (10^{-3}; 10^{-4} with $3\times$ targets as the CIFAR-100 rescue) and the controller (5) at $\rho = 0.25$ everywhere (drive form; a gradient-referenced form behaves equivalently, Appendix I). Full system in every cell; CIFAR-100 numbers are floor-regime and reported for ordering only (BP+ER 6–10%).

	fixed λ (tuned)		controller	
	seq.	static	seq.	static
split-MNIST	$\mathbf{92.7 \pm 0.3}$	94.7 ± 0.3	91.7 ± 0.2	$\mathbf{96.1 \pm 0.2}$
split CIFAR-10	$\mathbf{29.5 \pm 0.8}$	27.6 ± 0.4	26.5 ± 1.3	$\mathbf{29.9 \pm 0.3}$
feature CIFAR-10	$\mathbf{41.5 \pm 1.1}$	41.8 ± 1.1	38.9 ± 2.0	$\mathbf{43.0 \pm 1.2}$
split CIFAR-100	1.4 ± 0.5	1.1 ± 0.1	$\mathbf{3.4 \pm 0.6}$	$\mathbf{3.7 \pm 0.3}$
+ features	1.2 ± 0.3	1.2 ± 0.2	$\mathbf{5.7 \pm 0.8}$	$\mathbf{6.0 \pm 0.7}$

Energy. Converted to spiking inference (T_s=8), the default system runs at 93.8% (rate accuracy 94.7%) for a proxy of 119 nJ per sample against 2461 nJ for dense rate inference (20.7×) and 526 nJ for event-driven rate inference (4.4×); accuracy is monotone in T_s, whereas the rotation-free control degrades beyond T_s=8 (94.6 → 90.1 at T_s=32), consistent with rotation shaping sparser codes.

5 Scaling the mechanism: self-referenced decay and depth

The last tuned constant. Every result above runs the learning rule with a decay λ tuned once per dataset (10^{-3} on the image benchmarks). Split CIFAR-100 exposes what that constant hides: with a hundred classes each class receives a tenth of the supervision events while the decay acts on every synapse at every step, so at the MNIST-tuned value the hidden weights follow the pure-decay trajectory $(1-\lambda)^t$ exactly, activity collapses and the network lands at chance (2.4% i.i.d. against 20–23% on a ten-class subset of the same images). The class count is a hidden variable every fixed decay is silently tuned against. We replace the constant by a per-layer ratio controller,

$$\lambda_\ell = \min\Big(\lambda_{\max},\ \rho\,\mathrm{EMA}\big[u^\ell\big]\big/\big(\overline{|W^\ell|} + \epsilon\big)\Big), \qquad \rho = 0.25,\ \lambda_{\max} = 0.02, \tag{5}$$

where u^ℓ is the mean magnitude of the data-driven, post-optimiser, *pre-decay* waking increment applied to layer ℓ (so the decay never feeds its own drive), the EMA (coefficient 0.02) runs over waking unmasked steps only, $\epsilon = 10^{-12}$ floors the denominator, and W^ℓ and $B^{\ell-1}$ share λ_ℓ so alignment is untouched: the decay removes a fixed fraction of what learning currently adds. ρ is the only control target; $\lambda_{\max}$, the EMA coefficient and ϵ are global safeguards never tuned per dataset. Isolation and rotation decide *where* and *when* to consolidate; the controller decides *how strongly* to update as the scale changes, by the principle behind every control signal in this paper — novelty as a fast-over-slow error ratio, homeostasis as a unit's own use, decay as a drive-over-weight ratio: all referenced to the learner's own current state, none absolute.

One ratio target, every dataset (Table 2). With the same ρ on every dataset the controller concedes 1–3 sequential points to the per-dataset tuned value and wins on every static axis by 0.5–3.6 — the fixed decay had under-regularised every thick-signal regime — with 96.1 ± 0.2 the best i.i.d. accuracy of the two-layer substrate under any configuration. On CIFAR-100 it lifts the system 2.6–5× over the manual rescue with no dataset-specific decay coefficient. The realised λ_ℓ self-differentiates as the tuning history would predict ($\sim 10^{-5}$ on MNIST sequential streams, $\sim 10^{-4}$ on static ones, up to 3×10^{-3} on raw CIFAR): the per-dataset ladder was a signal-to-decay ratio in disguise. The controller must see only unamplified waking updates, and it does not subsume λ's role as a capacity regulariser on small tabular problems (Appendix I).

Depth (Figure 3). Depth attenuates the error signal per layer as the class count thins it per class. Under the tuned fixed decay the static axis dies at four hidden layers (9.7 ± 0.7%) and the sequential axis turns bimodal at five (55.5 ± 39.3); under the controller every cell stays alive (90.6 → 90.0 → 87.6 sequential, 95.6 → 95.3 → 91.8 static). The residual toll is architectural: ResNet-style skip synapses — S^ℓ carrying $a^{\ell-2}$ into each deep hidden layer, learned by the same local product with a Kolen–Pollack feedback twin and the shared decay, replay-masked under the same pre-or-post-asleep rule — restore the thick error path, and the five-hidden-layer system returns to the two-layer record

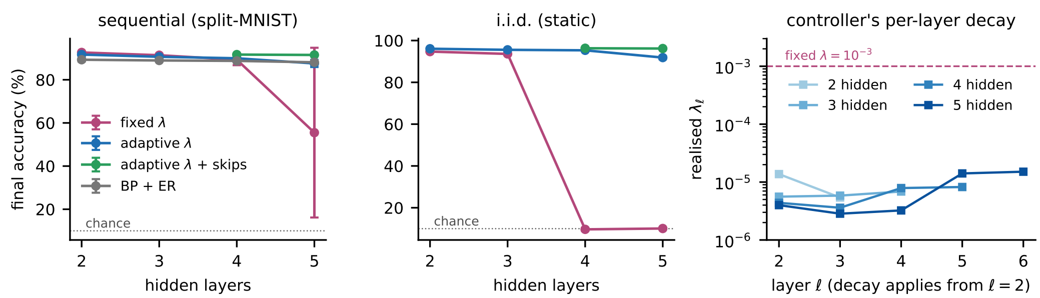

Figure 3: **Depth is the same starvation on another axis, and two mechanisms cross it.** Sequential (left) and static (middle) accuracy against hidden depth (three seeds): the tested fixed decay collapses, the controller degrades gracefully, and controller + locally learned skip synapses return the five-layer stack to the two-layer record; BP+ER is depth-insensitive but forgets 2–3× more. Right: the controller's realised per-layer decay.

on both axes ($91.5 \pm 0.4/96.2 \pm 0.3$; 96.3 ± 0.2 static at depth four). The bypass alone also rescues the fixed decay ($91.3/93.6$ at depth five), consistent with error-path attenuation being a principal contributor to the collapse; the controller adds its static margin on top. BP+ER at the same widths is depth-insensitive ($\approx$88–89) but forgets 2–3× more ($F \approx 13$–14 vs. 5–7); the local system with skips and the controller beats it at every depth tested. The two starvations are not interchangeable: on CIFAR-100 depth is a net loss that skips only half refund, while width pays (Appendix I).

6 Discussion

What this offers agents. The construction is a recipe rather than a benchmark result: if an agent's representation is sparse enough that a batch leaves most units silent, and its learning rule yields a local error signal, consolidation can run in the silent degrees of freedom *while the agent acts* — with an exactness guarantee on the computation in use, a homeostatic trigger that needs no task boundaries, and a replay budget an online system can afford. The study also says where the recipe breaks: unmasked optimiser state voids the guarantee, offline rehearsal and momentum-free updates lose the variance averaging isolated micro-batches need, and a fixed synaptic decay is a hidden per-dataset constant that depth and class count both expose. We treat the biology as motivating analogy, not mechanistic equivalence: the refractory rule mirrors the finding of Driessen et al. [2026] that ON/OFF alternation, not tonic reduction, discharges local sleep pressure — our soft-rotation control, a tonic reduction, is reliably worse — and the dissociation of triggers (homeostatic during wake, novelty-timed for offline rehearsal) is a testable prediction.

Limitations. All results are at MLP scale on MNIST/CIFAR variants with three to six seeds; every configuration choice during development was made on the official test sets, and Appendix J replicates the headline MNIST orderings on a single held-out split (mean gap 0.9 points). BP+ER keeps a 1.7-point lead on feature-based CIFAR-10, attributed by the tested decomposition to dense activation; rotation keeps a static price of about a point that no tested gate removes without a sequential cost; CIFAR-100 is floor-regime for every method; the energy figure is an arithmetic proxy. The primitives all have ancestors; the contribution is their combination, with an exactness argument and a measured boundary. Within these limits the conclusion stands: consolidation does not require an offline phase — the learner consolidates while it infers, above the strongest offline-rehearsal schedule at 2.4× its replay samples and within a point of it at a matched budget. Code will be released.

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A Proofs

Proof of Proposition 1. By induction on ℓ, for every sample b. Assume $a^{\ell-1}_{\cdot,b}$ is unchanged (true for $\ell = 1$ since $a^0_{\cdot,b} = x_b$). For any unit i, $\tilde{z}^{\ell}_{i,b} - z^{\ell}_{i,b} = \sum_j \Delta^{\ell}_{ij} a^{\ell-1}_{j,b}$, and each term vanishes: if $j \in \mathcal{A}^{\ell-1}(X)$ then $\Delta^{\ell}_{ij} = 0$, otherwise $a^{\ell-1}_{j,b} = 0$ for every b. Hence $z^{\ell}_{\cdot,b}$, and $a^{\ell}_{\cdot,b}$ through (1), are unchanged; the induction closes at the last hidden layer, and at the output layer whenever the readout row satisfies the same condition. □

Proof of Proposition 2. A unit outside $\mathcal{A}^{\ell}(X)$ contributes $a^{\ell}_{i,b} = 0$ on every sample. After the update its pre-activation on sample b is $z^{\ell}_{i,b} + \delta_{i,b}$; under the stated margin it is still excluded by Φ_k (or rectified to zero), so $a^{\ell}_{i,b} = 0$ still, every other unit's input is untouched by the argument of Proposition 1, and the layer's activities are identical. The readout row is excluded from both propositions in the default system because it is deliberately kept plastic. □

B The price of the substrate

Table 3 traces the path from a dense backpropagation MLP to the substrate used throughout, on i.i.d. MNIST. The cumulative ladder prices the entire constraint set at 1.2 points below the dense reference for $4\times$ fewer synaptic operations per pass; the local burst-error objective itself costs nothing. Settling cannot merely be shortened (T=1 collapses); the single-phase burst sweep matches five-step settling in one pass. The Kolen–Pollack decay and even restored weight transport are accuracy-neutral at this scale — the decay earns its keep as a regulariser on harder inputs. Spiking-flavoured training fails outright: spiking is affordable at inference, not as a training-time code. On the plain substrate SGD loses 1.3 points to Adam, yet inside the consolidation loop it wins on both axes: per-synapse adaptive state is an asset while every synapse trains on every batch and a liability once updates become mask-confined.

Table 3: **Substrate ablation** on i.i.d. MNIST (256–128 hidden core, three seeds). "Syn. ops" is the fraction of the dense MLP's synaptic operations per forward pass. Upper block cumulative; lower block replaces one ingredient at its stage.

Substrate	Accuracy	Syn. ops
dense backpropagation MLP (autograd reference)	97.7 ± 0.1	1.00
local burst-error learner, unconstrained	97.8 ± 0.1	0.85
+ bounded error, Dale's law, sign-concordant feedback (no transport)	97.1 ± 0.0	0.85
+ k-WTA 10%	97.0 ± 0.2	0.80
+ relaxation T=5	97.0 ± 0.1	0.80
+ 30% distance-dependent connectivity	96.5 ± 0.2	0.24
+ burst gate and burst baseline	96.3 ± 0.2	0.24
+ single-phase burst sweep replacing relaxation (*final substrate*)	96.5 ± 0.1	0.24
random wiring instead of distance-dependent, equal density	95.4 ± 0.0	0.24
one settling step (T=1) instead of five	17.9 ± 6.8	0.80
weight transport restored (mirroring, no KP decay)	96.3 ± 0.2	0.24
Kolen–Pollack decay removed	96.3 ± 0.1	0.24
heavy-ball SGD instead of Adam ($\eta = 0.02$)	95.2 ± 0.3	0.24
k-WTA 5% / 2% instead of 10%	95.9 ± 0.1 / 94.2 ± 0.2	0.79/0.78
multiplicative burst coding	78.8 ± 1.9	0.24
training on T_s=8 spike counts	80.0 ± 0.8	0.24

C Protocol details, cost accounting and the operating frontier

The V1-like front-end consists of 256 normalised random $6\times6\times3$ patches sampled from the training images, applied with stride 2, rectified against each filter's mean response over training images, and quadrant-average-pooled (1024-D); nothing in it is learned. Split CIFAR-10 uses 32×32 luminance so that distance-dependent wiring keeps its image geometry. The buffer uses reservoir writing; the night reference uses a 256–128 core, its preferred width in our sweeps. Runs are deterministic at a fixed CPU thread count.

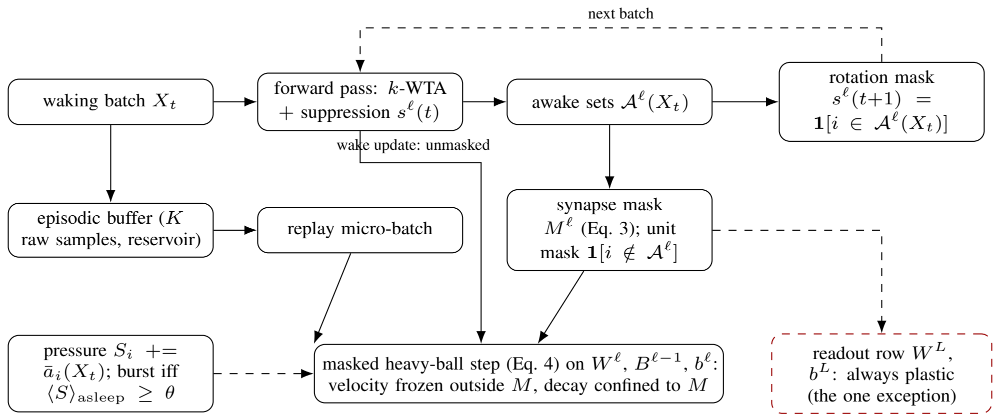

Figure 4: **Training data flow.** Each waking batch is inferred under the current suppression mask; its awake sets define the synapse and unit masks for the replay micro-batch applied in the same step and the rotation mask for the next batch. The masked step confines the velocity and the decay to the mask; the readout row is the one deliberately plastic exception. Unit-level pressure decides when replay bursts run.

Replay cost is reported in samples (R = replay batches $\times$ batch size). The night costs $480 \times 256 \approx 1.2 \times 10^5$ samples per run. The headline configuration (one 16-sample micro-batch per waking batch) costs $18{,}760 \times 16 \approx 3.0 \times 10^5$ ($2.4\times$ the night) for 92.7 ± 0.3. Two schedules meet the night's budget within $1.2\times$: eight-sample micro-batches at every waking batch (1.5×10^5, 91.8 ± 0.3) and 16-sample micro-batches at every second one (1.5×10^5, 91.7 ± 1.0 sequential, 93.5 ± 0.4 static); pressure-triggered bursts with batch 64 cost 1.8×10^5 ($1.4\times$) for 89.1 ± 1.1, and halving the cadence with doubled batches costs 1.3 points (91.4 ± 0.3). Accuracy is bought by *frequency* of small isolated updates rather than by replayed samples. Figure 5 draws the operating frontier of the record substrate.

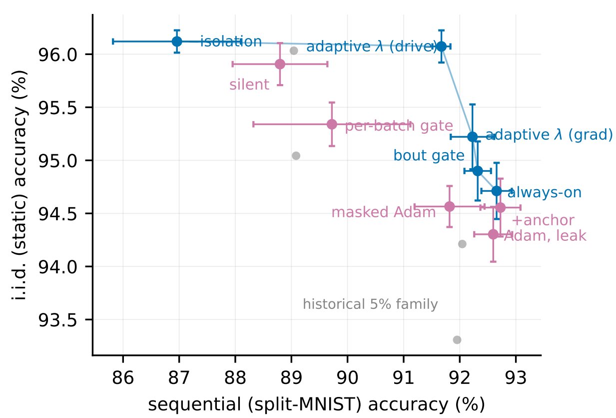

Figure 5: **Operating frontier** at the 10% activity fraction under heavy-ball SGD: split-MNIST sequential accuracy against i.i.d. static accuracy. Blue: non-dominated configurations; purple: dominated; grey: the 5% family of earlier phases. No single configuration wins both axes; always-on rotation is the sequential default and the adaptive decay extends the static side.

Baseline tuning and paired comparison. The offline-night reference was swept over replay batch count (20, 40 per night), replay batch size (16, 64, 256), learning-rate gain ($3\times$), core width (256–128 and 512–256), buffer size (200, 1000, 5000) and timing (clocked after each epoch, novelty-timed, surprise-timed), and its best sequential cell is reported; BP+ER was swept over learning rate ($3{\cdot}10^{-4}$,

10^{-3}, $3{\cdot}10^{-3}$), width and buffer size; the local learner over η (Appendix G), replay batch size, cadence and the gates of Table 1. Selection everywhere was sequential accuracy at $K{=}1000$ with the static axis reported alongside, on the official test set (Appendix J). Pairing seeds by index, the local system's margin over the narrow-substrate night is $+2.1$ points (bootstrap 95% CI $[+1.4, +2.5]$, three seed pairs; Welch $p = 0.004$), over the same-substrate night $+1.7$ ($[+0.1, +4.6]$, six pairs; the night's seed variance makes this one inconclusive, $p = 0.25$), over unmasked interleaved ER $+5.8$ ($[+5.0, +7.0]$) and over BP+ER $+3.5$ ($[+2.7, +4.2]$).

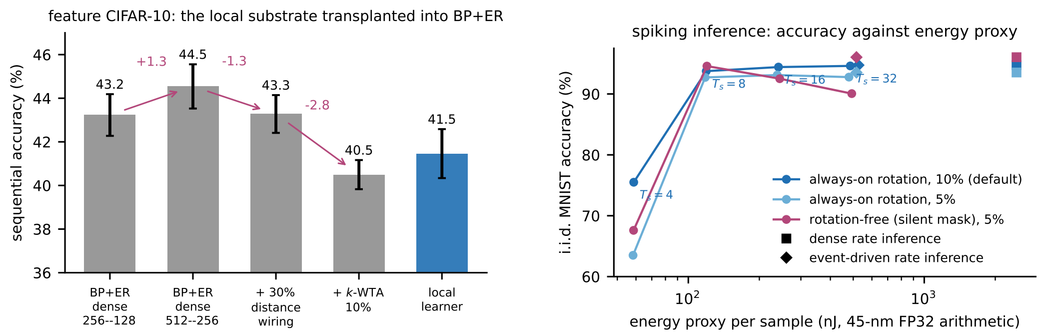

Figure 6: **Left:** the substrate decomposition on feature CIFAR-10 — transplanting the local network's ingredients into BP+ER one at a time (each cell at its better of two learning rates; six seeds except the local learner, three). **Right:** spiking inference, i.i.d. MNIST accuracy against the idealised 45-nm FP32 arithmetic-energy proxy for $T_s \in \{4, 8, 16, 32\}$, with the dense (square) and event-driven (diamond) rate references; rotation-trained networks are monotone in T_s, the rotation-free control degrades.

D Rotation gates at the 5% activity fraction

At the 5% fraction of the earlier phases, always-on rotation cost 1.3–3.0 points on i.i.d. MNIST ($95.0 \to 92.0$–93.7). Gating rotation by unit-level pressure only trades the axes (static $91.6 \to 94.0$ as sequential $88.8 \to 80.7$). The relative-novelty gate removes the price at no sequential cost on MNIST (static 95.0 ± 0.4 with rotation on 1.4–17% of batches; sequential 90.3–90.9). On split CIFAR-10 the price reverses — rotation gains $+2.5$ to $+6.5$ static points in the underfit regime — while the pure novelty gate misfires there (sequential 19.1 ± 1.1: fast/slow $\to 1$ on a stream that never becomes predictable), which the unmastered clause resolves with one setting across all four regimes (27.2 ± 0.7 / 25.3 ± 1.0 on CIFAR, 90.4 ± 0.9 / 94.3 ± 0.2 on MNIST). Those gate numbers use masked Adam; under heavy-ball SGD the per-batch gate's flicker destroys the optimiser's gain and the gate must be committed to minimum-duration bouts ($M{=}2048$ batches, loosely modelled on the consolidated bouts of the sleep–wake switch [Saper et al., 2005]; Table 1). Figure 7 shows the three regimes.

E Probes of the residual static price

Bout length and the OFF-side refractory. Bout length is a proper knob: $M{=}512$ gives $91.7 \pm 0.4/94.0 {\pm} 0.5$, recovering most of what the per-batch gate loses, while $M{=}4096$ drives the static duty to 0.94 and the refund vanishes. Completing the switch with an OFF-side refractory (triggers ignored for M_{off} batches after a bout) is a clean null: short refractories never engage on the static stream, and a long one (4096) lowers the static duty to 0.53 without moving static accuracy (94.0 ± 0.7) while monotonically eroding the sequential axis ($92.1 \to 91.0$). The residual static price sits in the rotation of settled coalitions itself, not in how often the gate fires.

Structural exemption and synaptic anchoring. Sparing the top-q units by long-term use from ever being benched traces a smooth trade-off ($q{=}0.10$: $90.9 \pm 0.1/94.2 \pm 0.2$; $q{=}0.25$: $89.1 \pm 0.4/95.4 \pm 0.2$) that the bout gate dominates: temporal commitment beats structural exemption. Two-timescale synapses in the spirit of synaptic consolidation cascades [Benna and Fusi, 2016] — a slow anchor

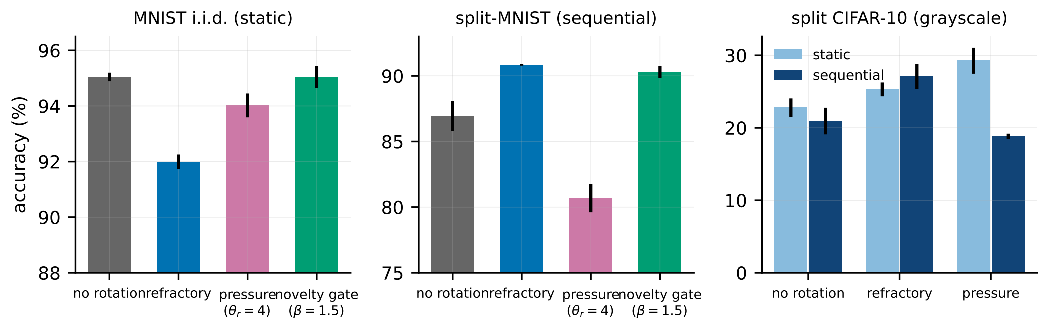

Figure 7: **Rotation at 5%: price, trade-off and gate.** Left: on i.i.d. MNIST the novelty gate restores no-rotation accuracy. Middle: on split-MNIST rotation is required and the gate keeps it. Right: on split CIFAR-10 rotation is a gain on both axes.

absorbing the fast weight at rate μ while the fast weight decays toward it at rate λ, ticking only on waking updates — do not dissolve the price (no cell exceeds the anchor-free static accuracy); weak symmetric coupling acts as a sequential stabiliser at 5% (92.4 ± 0.2, variance halved) and is neutral at 10% ($92.7 \pm 0.4/94.6 \pm 0.3$).

F Small buffers and sleep-anchored down-selection

At K=200 the night appeared to keep a 2.6-point edge; on the same 512–$256/5\%$ substrate the night reaches 84.0 ± 3.4 while noisy pressure-burst local sleep reaches 84.3 ± 1.0 and local+night 84.8 ± 0.9 — parity, with three-fold smaller variance for the wake-side scheme. The strongest tiny-buffer number remains the narrower night (86.9), so the offline window retains an advantage only in combination with narrow substrates. Replay diversity (input noise σ=0.2) is a wake-side lever — it lifts every local scheme ($81.8 \pm 2.9 \rightarrow 84.3 \pm 1.0$) and hurts the night — and unmasked waking replay collapses (71.0 ± 1.7). Anchoring structural plasticity (prune the weakest live synapses, regrow with a distance bias) to consolidation windows, motivated by synaptic homeostasis [Tononi and Cirelli, 2014], is free on the continuous substrate (90.9 ± 0.2 vs. 90.8 ± 0.1), rescues the wide-sparse night (87.2 ± 3.8 vs. 84.1 ± 1.3), and harms the episodic-burst substrate (84.6 ± 0.5 vs. 89.9 ± 0.1): down-selection, like replay, needs either density or an offline window.

G Learning-rate sensitivity

At the 5% fraction, under always-on rotation, SGD holds a sequential plateau over $\eta \in [0.01, 0.02]$ (91.9 ± 0.5/92.0 ± 0.6) and a static plateau over $[0.02, 0.05]$ (93.3–93.7), decaying gracefully outside (86.3/89.2 at η=0.1); masked Adam, swept over a tenfold range, has its sequential optimum at the default (90.8 ± 0.1 at 10^{-3}; 88.6 at $3{\cdot}10^{-4}$, 88.9 at $3{\cdot}10^{-3}$), and no Adam setting reaches the heavy-ball joint point. Unmasked SGD is poor and unstable at every rate (84–88, seed deviations 1.8–4.1). The same two-sided sweep at the 10% record fraction (three seeds) gives the same verdict: heavy-ball SGD holds $92.2 \pm 0.7/93.3 \pm 0.2$ at η=0.01, $92.7 \pm 0.3/94.7 \pm 0.3$ at 0.02 (the joint optimum) and $91.5 \pm 0.6/94.7 \pm 0.3$ at 0.05, decaying at 0.005 (91.3/91.3) and destabilising at 0.1 (79.5 ± 17.3); masked Adam has its optimum at the default ($91.8 \pm 0.6/94.6 \pm 0.2$ at 10^{-3}; $89.9/89.9$ at $3{\cdot}10^{-4}$, $90.9/94.3$ at $3{\cdot}10^{-3}$). The optimiser comparison of Table 1 is made at each optimiser's best rate.

H Momentum, exactness and other negative results

Momentum. With momentum zero the default replay step ($3\times$ the waking rate) diverges or starves hidden activity at every learning rate tested ($\eta \in [0.02, 0.4]$: $\approx 10\%$, hidden dimensionality 0); the waking path alone learns normally at the matched effective step, so the collapse is the replay path's: the velocity's $\sim 10\times$ averaging of the 16-sample replay micro-batches is what heavy-ball provides.

Re-scaling the replay gain (η=0.2, gain 0.3) makes pure SGD viable at $89.9 \pm 0.3/90.1 \pm 0.2$ ($91.7 \pm 0.2/91.4 \pm 0.0$ with the adaptive decay); other settings are fragile.

Exactness measurement. Under the 5% Adam-era configuration the isolated update altered the top hidden code of 1.0% of waking samples per step (against 0.36% for the record configuration); the numbers in Section 3 are population rates over every replay step of a full run.

Falsified under matched budgets. A frozen dentate-gyrus-style sparse expansion front-end (input decorrelation does not propagate through learned k-WTA layers: input task overlap drops $0.79 \rightarrow 0.18$ yet every downstream metric worsens by 3–11 points); generative (REM) replay as a substitute for episodic samples; margin-based reverse learning; surprise-gated memory writing; multiplicative burst coding; absolute-threshold rotation gates; soft (gain-reduction) rotation; mirror-direction isolation; OFF-side gate refractories; strongly coupled synaptic anchors; utility-exempt rotation.

I The controller: composition rules, CIFAR-100 and reproducibility

A gradient-referenced form of the controller ($\rho\,\eta\,\mathrm{EMA}[\overline{|g^\ell|}]/\overline{|W^\ell|}$) behaves equivalently (split-MNIST $92.2 \pm 0.4/95.2 \pm 0.3$, six seeds; CIFAR-10 $28.1 \pm 0.9/31.2 \pm 0.8$; feature CIFAR-10 $40.5\pm0.6/43.8\pm0.2$; CIFAR-100 $3.7\pm0.4/4.3\pm0.2$). The drive estimate must see only unamplified waking updates: inflated one-hot targets and the offline night's $3\times$-gain batches push the EMA to its clip and erase day learning (night $\times$ controller 60.2 ± 2.8 against 90.9 ± 3.2 for the night alone), whereas anchor and bout gate compose cleanly (bout + controller 96.0 ± 0.3 static). The controller does not subsume λ's role as a capacity regulariser against overfitting on small tabular problems, information no drive ratio contains. On CIFAR-100 (ten tasks of ten classes; every method at floor, chance 1%, BP+ER 9.9 ± 0.5 with twice the local learner's forgetting) depth is a net loss that skips only half refund (four hidden layers $1.9/1.6 \rightarrow 3.8/3.0$ with skips against $5.7/6.0$ at two), while width at the short chain length lifts the local record to $6.7 \pm 0.3/7.5 \pm 0.2$ (1024–512; a $5\times$ buffer adds nothing): the thin-signal regime wants capacity, not depth, and the static-axis wall (6–7.5$\%$ under every substrate change) says the front-end is the binding constraint there. All runs reproduce bitwise at a fixed CPU thread count; a different thread count changes the reduction order, which k-WTA's discontinuity amplifies into a different trajectory.

J Held-out replication of the headline orderings

Every configuration choice made during development — some four hundred sequential-axis configurations — was made on the official test sets. As a check on the orderings that carry the claims, the headline configurations were re-run with a tenth of the training set held out (stratified, fixed split, never trained on) as the evaluation set, three seeds each (Table 4). Held-out numbers sit 0.9 points below test on average, the sequential-axis ranking correlates at Spearman $\rho = 0.97$, and the headline MNIST orderings survive: always-on rotation and the bout gate lead (91.6 ± 0.3/91.8 ± 0.3 vs. the night's 89.0 ± 3.5, BP+ER's 88.8 ± 0.3 and unmasked ER's 85.4 ± 3.5; paired bootstrap margins $+2.6$ $[+0.2, +5.2]$, $+2.8$ $[+2.5, +3.2]$ and $+6.3$ $[+3.8, +9.1]$ over six seed pairs); the silent mask trails rotation; the five-layer skip system holds the two-layer level on both axes (91.0/95.8). One claim does not replicate cleanly: the adaptive decay's static-side margin (96.1 vs. 94.7 on test) becomes 93.5 ± 4.3 vs. 94.1 ± 0.2 on the held-out split, where one of six seeds collapsed — we report it as unconfirmed. This is a single fixed split of MNIST only.

Table 4: **Held-out replication.** Sequential / static accuracy on the official test set (all available seeds, three to six) and on a held-out tenth of the training set (six seeds).

	sequential		static	
	test	held-out	test	held-out
always-on rotation (default)	92.7 ± 0.3	91.6 ± 0.3	94.7 ± 0.3	94.1 ± 0.2
bout-committed gate	92.3 ± 0.2	91.8 ± 0.3	—	—
adaptive λ (drive)	91.7 ± 0.2	90.6 ± 0.6	96.1 ± 0.2	93.5 ± 4.3
five hidden layers + skips + adaptive λ	91.5 ± 0.4	91.0 ± 0.3	96.2 ± 0.3	95.8 ± 0.3
silent mask (no rotation)	88.8 ± 0.8	88.0 ± 0.7	—	—
offline night, same substrate	90.9 ± 3.2	89.0 ± 3.5	—	—
unmasked interleaved ER	87.0 ± 1.1	85.4 ± 3.5	—	—
BP + ER	89.3 ± 0.8	88.8 ± 0.3	—	—
no buffer	19.6 ± 0.0	19.4 ± 0.0	—	—